Tools Saved 30 Jun 2026 at 10:41 Published

Preview Publish

Question type

Graphical Perception (Position-Angle)

ExpertReview score Fair

Text entry

Prolific ID

Prolific ID

Your Prolific ID should automatically appear here. If it does not appear, please provide your Prolific ID below.

[redacted]

Import from library Add new question

Add validation

Add Block

Info & Consent

PIS 1

Participant Information Page

REC reference number (ETH2526-1929), 21/5/26

Title of research
Graphical Perception of Position and Angle: A Replication and Follow-up of *Cleveland & McGill (1984)*

Name of principal investigator/researcher: Idowu Afolabi Isaac Ojo

Invitation paragraph
We would like to invite you to take part in our research study. Before you take part, we would like you to understand why the research is being done and what it would involve. Please read the following information carefully.

PIS 2

What is the purpose of the research?
This study investigates how accurately people estimate quantities from different types of charts. It compares visual estimates made from bar charts and pie charts. This study is a replication and follow-up of a classic study in graphical perception by *Cleveland and McGill (1984).*

The original work tested how accurately people estimated one chart value as a percentage of another chart value. Our study will repeat this type of task and include some others. This will help us understand whether bar charts and pie charts support different comparison tasks equally well.

The study is being conducted as part of a PhD Data Visualization research project at City St George's and the University of Warwick.

The online task is expected to take 20 minutes on average and 30 minutes maximum.

PIS 3

Why have I been invited to take part?
You have been invited because you are registered on Prolific and meet the study eligibility criteria.
To take part, you must:

- be aged 18 or over;
- be fluent in English;
- have normal or corrected-to-normal vision;
- use a desktop or laptop computer rather than a mobile phone or tablet;
- be able to complete a short online visual judgement task.

PIS 4

Do I have to take part?
Participation in the project is completely voluntary, and you are free to choose not to complete the study. You can stop participating at any point during the survey by exiting the webpage. It is up to you to decide whether to take part or not. If you decide to take part, you are still free to stop participating at any time without giving a reason. If you withdraw before you have submitted responses, your data will not be used in the final analysis. All data is pseudonymized, which means the data will not be fully anonymous, as we will collect your Prolific IDs for payment and withdrawal purposes. This means that you are free to withdraw from the study before you have completed and clicked to submit your responses, but not once you have done so.

PIS 5

What will happen if I take part?
If, once you have read this information, you agree to take part, you will access the study and be asked to complete it online. You will first see the study information and consent questions.

During the task you will:

- complete practice trials so you understand the task;
- view a series of bar charts and pie charts;
- make quick visual estimates by entering percentage values

You will be asked to make a reasonable visual estimate for each question. You should try to answer as accurately as you can using only your visual judgement, but you should not use calculators, rulers, screenshots, note-taking, marks on the screen, or any other external tools.

If you do not want to take part and do not consent, simply close this browser.

PIS 6

What are the benefits and risks of taking part?

This is a low-risk study - the main possible burdens are mild eye strain, fatigue, or frustration from making visual estimates. To reduce these risks, the study is kept short, designed to take approximately 20 minutes on average, and no longer than 30 minutes. You will have the option to take a short break halfway through the study at your own discretion.

We are interested in how people interpret charts visually, not in assessing or scoring individual participants. Your participation will help us understand how people interpret common chart types and may contribute to improved data visualization design and research methods.

PIS 7

What data will be collected?
We will collect the minimum data needed to run and analyse the study. This may include:
• your date of completion;
• your approximate location (longitude-latitude);
• study metadata such as, completion status, condition assignment, completion time;
• your consent response;
• your responses to the chart judgement questions;
• your Prolific ID, used temporarily for payment administration and withdrawal purposes.

Payment will be facilitated through Prolific. The researchers will not have access to your financial information. The dataset will be fully anonymised by removing Prolific IDs once payments have been made. The fully anonymised dataset, with Prolific IDs removed, will be made publicly available through a research repository. Collaborators from the University of Warwick may access fully anonymised research data only.

Q28

What will happen to the results?
The results may be used in a research report, cohort presentation to academics within the institution, conference, publication, or future research output.

PIS 8

Who has reviewed the research project?
This research project (ETH2526-1929) has been approved by the Computer Science Research Ethics Committee of City St George's, University of London.

PIS 9

Data privacy statement
City St George's, University of London is the sponsor, and the data controller of this study based in the United Kingdom. This means that we are responsible for looking after your information and using it properly. The legal basis under which your data will be processed is City St George's public task.

Your rights to access, change or move your information are limited, as we need to manage your information in a specific way for the research to be reliable and accurate. To safeguard your rights, we will use the minimum personally identifiable information possible (for further information please see https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/a-guide-to-lawful-basis/public-task/). The only people at City St George's who will have access to your identifiable information will be members of the research team, and, if appropriate, individuals with responsibility for monitoring and auditing at City St George's, including of research projects. There may be occasions when regulatory authorities may access research data in accordance with their statutory powers.

You can find out more about how City St George's handles personal data by visiting https://www.city.ac.uk/about/governance/policies/data-protection-policy. You can also read City's general privacy notice by visiting https://www.city.ac.uk/about/governance/policies/general-privacy-notice . If you are concerned about how we have processed your personal data, you can contact the Information Commissioner's Office (ICO) directly https://ico.org.uk/.

PIS 10

Insurance
City St George's, University of London holds insurance policies which apply to this project, subject to the terms and conditions of the policy. If you feel you have been harmed or injured by taking part in this project, you may be eligible to claim compensation. This does not affect your legal rights to seek compensation. If you are harmed due to someone's negligence, then you may have grounds for legal action.

PIS 11

What if there is a problem?
If you have any problems, concerns, or questions about this project, you can speak to a member of the research team. If you remain unhappy and wish to complain formally, you can do this through the University complaints procedure. To complain about the project, you can write to or email the Secretary to Senate Research Ethics Committee and inform them that the name of the project is *Graphical Perception of Position and Angle: A Replication and Follow-up of Cleveland & McGill (1984).*

Research Governance, Ethics, and Integrity Manager
Research & Innovation Directorate
City St George's, University of London
Northampton Square, London, EC1V 0HB
Email: senaterec@city.ac.uk

PIS 12

Further information and contact details
If you have any questions or would like more information about this study, please contact a member of the research team at City St George's:

Sabah Azhar (research student): sabah.azhar@city.ac.uk

Idowu Afolabi Isaac Ojo (research student): idowu-afolabi.ojo@city.ac.uk

Prof Jason Dykes (academic lead): j.dykes@city.ac.uk

PIS 13

Thank you for taking the time to read this information sheet.

Page Break

Consent

Consent Page
If, once you have read this information, you agree to take part, please confirm that you have read, understood and agree with the following statements. You will then access the study immediately and be asked to complete it online.

- I confirm that I have read and understood the Information Sheet and been given the opportunity to contact the research team to ask questions;
- I understand that all personal information I provide will remain confidential and, where possible, efforts will be made to ensure that I cannot be identified;
- I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason and that closing the browser before I have submitted my results, will end my participation and remove my data from the study;
- I agree to take part in this study.

If you do not want to take part and do not consent, simply close your browser or select the 'I do not wish to participate in this study' option below where you will be redirected to Prolific.

○ I have read, understood and agree with the above statements and wish to participate in this study

○ I do not wish to participate in this study

Import from library | Add new question

Add Block

Example Instructions

Example Instructions

Example

You will now see an example question explaining how to interpret the study questions.

The example will include both a bar chart and a pie chart.

Import from library | Add new question

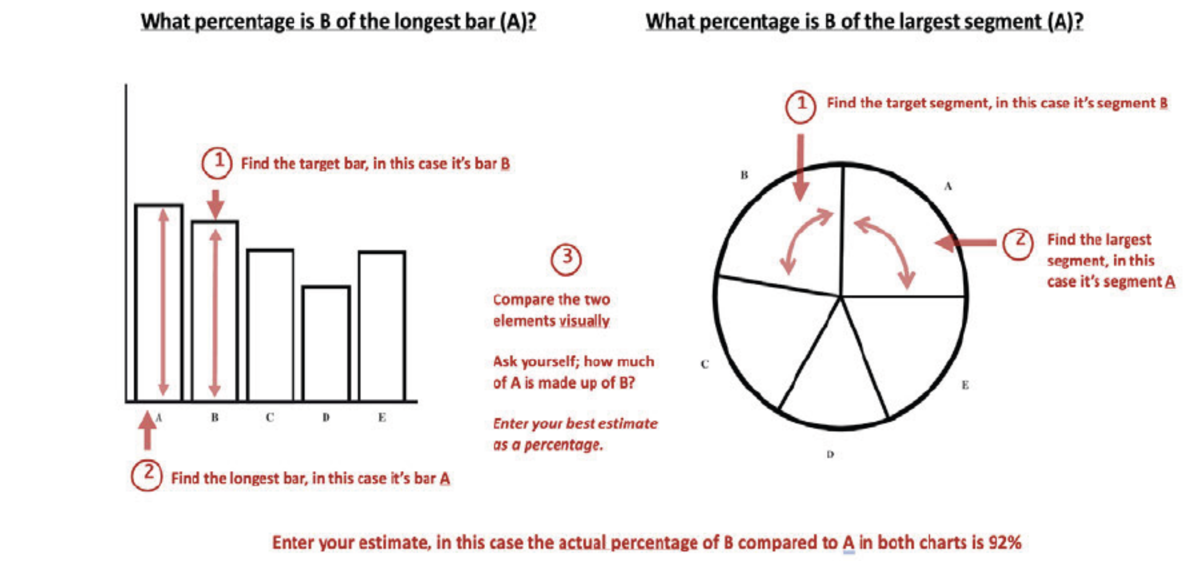
Add Block

Example - Part-Part

E1

Example

Please ensure you have understood the example below before continuing.



Import from library

Add new question

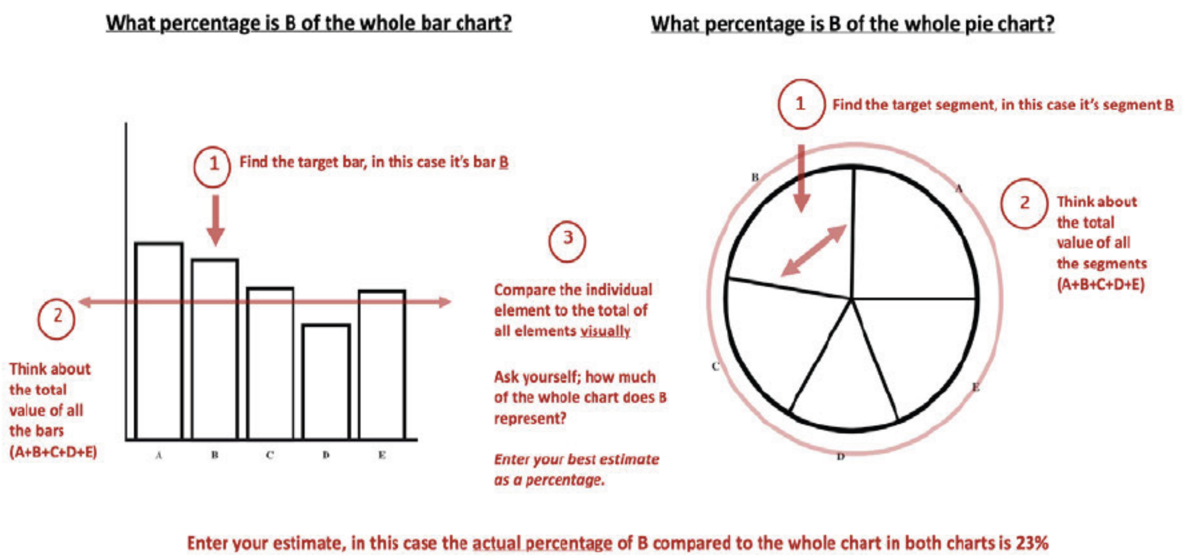
Add Block

Example - Part-Whole

E2

Example

Please ensure you have understood the example below before continuing.



Import from library | Add new question

Add Block

Practice Instructions

Practice Instruction

Practice Task

You will now complete 2 practice questions.

You will be shown a sequence of charts and asked to make a reasonable visual estimate about the size of their components.

Please write your answer in the text box **as a percentage in numerical form only** (i.e., 23, 47). **Do not include the '%' symbol in your answer.**

Do not use calculators, rulers, screenshots, note-taking, or external tools.

Exact accuracy is not expected.
We are interested in how people interpret charts visually, not in assessing or scoring individual participants.

Import from library | Add new question

Add Block

Practice - Part-Part

QP1

?

Import from library | Add new question

Add Block

Practice - Part-Whole

QP2

?

Import from library | Add new question

Add Block

Interim 1

Interim 1

You have completed the practice task.

Please press 'Next page' when you are ready to proceed to the main task.

Import from library | Add new question

Add Block

Main Instructions

Main Instructions

Main Task

You will now complete the main task.

You will be shown a sequence of charts and asked to make a reasonable visual estimate about the size of their components.

Please write your answer in the text box **as a percentage in numerical form only** (i.e., 23, 47). **Do not include the '%' symbol in your answer.**

Do not use calculators, rulers, screenshots, note-taking, or external tools.

Exact accuracy is not expected.
We are interested in how people interpret charts visually, not in assessing or scoring individual participants.

Import from library | Add new question

Add Block

Main - Part-Part

Q1

Import from library | Add new question

Add Block

Main - Part-Whole

Q2

Import from library
Add new question

Add Block

Interim 2

Interim 2

You have completed the main task.

Please press 'Next page' when you are ready to proceed to the debrief.

Import from library
Add new question

Add Block

Debrief

Debrief

Debrief

Thank you for participating in this study.

Please read the following information to learn more about the nature and purpose of the study. If you wish to withdraw at this point, please close this browser, or proceed to the next page where you will be able to make your decision about providing consent, if you choose not to provide consent, you will be redirected to Prolific.

Title of Experiment: Graphical Perception of Position and Angle: A Replication and Follow-up of *Cleveland & McGill (1984)*

Names of Researchers: Sabah Azhar, Fiona Chung, Idowu Afolabi Isaac Ojo, Cagatay Turkay, Jason Dykes

Ethics Approval Code: ETH2526-1929

This study is a replication and follow-up of a classic study in graphical perception by *Cleveland & McGill (1984)*. The original paper claims that judgements of position along a common scale (i.e., when examining bar charts) are more accurate than judgements of angle (i.e., when examining pie charts), however, the experiment only involved part-to-part comparison tasks.

Therefore, we aim to:

1. Replicate the Position-Angle Experiment reported in *Cleveland & McGill (1984)*, which compared estimation accuracy for bar charts and pie charts during part-to-part comparison tasks.
2. Extend this experiment in a follow-up study where we introduce a part-to-whole comparison task that was not included in the original study.

You will have completed both position and angle judgements for either the part-to-part comparison task or the part-to-whole comparison task.

If you have any questions or would like more information about this study, please contact a member of the research team at City St George's:

- **Sabah Azhar** (research student): sabah.azhar@city.ac.uk
- **Idowu Afolabi Isaac Ojo** (research student): idowu-afolabi.ojo@city.ac.uk
- **Prof Jason Dykes** (academic lead): j.dykes@city.ac.uk

Thank you again for your time. Please press 'Next page' when you are ready to proceed to the final consent form.

Cleveland, W. S., & McGill, R. (1984). Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. Journal of the American Statistical Association, 79(387), 531–554. https://doi.org/10.2307/2288400

Import from library

Add new question

Add Block

Final Consent

Final Consent

Final Consent Form

If you are happy for your data to be collected and analysed for this study, please confirm your final consent below.

If you do not wish to consent, you can close this browser or select the option below and you will be redirected to Prolific.

- I confirm that I am happy for my data to be collected and analysed for this study
- I do not wish for my data to be collected or analysed for this study

Import from library | Add new question

Add Block

End of survey message

End

Thank you for taking part in this study.

Please click the button below to be redirected back to Prolific to register your submission.

Import from library | Add new question

Add Block

End of Survey

We thank you for your time spent taking this survey.

Your response has been recorded.